No Free Depth: Exhaustive Layer-Duplication Scans Reveal Depth-Structured Robustness but No Reasoning Circuits in Large Vision Transformers

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Abstract

Recent work on large language models (LLMs) showed that re-executing a contiguous block of transformer layers at inference time—without any weight modification—can *improve* downstream performance, and attributed this to "reasoning circuits" occupying a format-agnostic middle region of the network. We ask whether this phenomenon transfers to vision. We exhaustively scan all $L(L+1)/2$ contiguous layer-duplication configurations in the two largest openly available pure vision transformers spanning both dominant training regimes—EVA-CLIP-18B (language-supervised, 48 layers, 1,176 configurations) and DINOv3-7B (self-supervised, 40 layers, 820 configurations)—scoring each configuration with training-free nearest-centroid probes over eleven datasets that span six real-world domains and five procedurally generated visual-reasoning tasks. A selection-robust two-stage protocol (borderline and random test splits, matched noise nulls, per-image bootstrap, and confirmation of top configurations against random-configuration controls on held-out images) yields a clear negative: *no* duplication configuration in either model produces gains that survive confirmation. The manner of failure, however, is highly structured and informative. Early-layer duplication is universally destructive; late-layer duplication is essentially free—the reverse of the LLM finding. A pre-registered representational-anatomy experiment explains part of this structure: a content-dominant layer band exists only in the self-supervised model, where it extends to the final layer, consistent with vision encoders lacking the "decode" phase whose disruption penalizes late-layer duplication in LLMs. Both models nonetheless exhibit small ($+0.5$–$2\,\mathrm{pp}$) but statistically reliable second-pass benefits in their mid-to-late stack. We release all code, per-image outcomes, and analysis artifacts.

1 Introduction

A transformer's forward pass applies a fixed sequence of residual blocks exactly once. Hankins [Hankins, 2025a] demonstrated that this convention can be profitably violated in large language models: executing a well-chosen contiguous block of middle layers *twice* at inference time—reusing the same weights, with no training—improved a 72B-parameter model's average score across six public benchmarks, with gains up to $+17.7\,\mathrm{pp}$ on individual reasoning tasks. Follow-up work generalized the effect across model families [Hankins, 2025b] and proposed a mechanism [Hankins, 2025c]:

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middle LLM layers operate on a shared, format-agnostic semantic representation—demonstrated by cross-lingual similarity analysis—so a middle layer's output is a valid input to the same region, and a second pass acts as additional computation in a stable representational space. Early layers (which map tokens into that space) and late layers (which map it back to token space) tolerate no such recycling.

Whether any of this transfers to vision is unknown, and the question is sharper than it may first appear. Vision transformers [Dosovitskiy et al., 2021] share the residual architecture that makes duplication mechanically possible, and prior work has found substantial robustness of both CNNs and LLMs to layer-level perturbation [Huang et al., 2016, Lad et al., 2024]. But two structural properties differ. First, a vision *encoder*'s output is already an embedding: there is no analogue of the return to token space, so the LLM three-phase picture (encode–reason–decode) may be missing its third phase entirely. Second, the representational content of a vision model depends on its training objective in a way with no LLM parallel: a language-supervised model (CLIP-style [Radford et al., 2021]) is optimized to preserve exactly the appearance attributes—color, style, background—that a "format-agnostic content space" would be expected to discard, whereas a self-supervised model faces no such pressure.

We address the question with what is, to our knowledge, the first exhaustive layer-duplication study in vision, designed around three methodological commitments:

1. **Exhaustiveness and scale.** We scan *every* contiguous duplication configuration in the largest openly available pure vision transformer, EVA-CLIP-18B [Sun et al., 2024] (17.5B parameters, 48 layers, 1,176 configurations), and repeat the identical experiment on the largest openly available self-supervised vision transformer, DINOv3-7B [Siméoni et al., 2025] (40 layers, 820 configurations), making the training objective the only varied factor.
2. **Selection-robust inference.** Ranking 1,996 configurations on small probe sets guarantees an inflated maximum (the winner's curse). Our protocol separates *screening* from *confirmation*: top configurations are re-evaluated on never-before-used images, with resampled reference sets, against the empirical distribution of *random* configurations, under a criterion fixed before the sweep. Complementary controls (a boundary-balanced test split, matched Gaussian-noise nulls, and per-image paired bootstrap) bound every source of optimistic bias we could identify.
3. **Registered mechanistic predictions.** Before either sweep completed, we ran a vision analogue of the cross-lingual anatomy experiment of Hankins [2025c]—an 8×8 shape-by-rendering-style stimulus grid replacing the original topic-by-language sentence grid—and recorded the predictions it makes about each model's duplication tolerance. The sweeps then graded those predictions.

Findings. The headline result is negative and unambiguous: in neither training regime does any duplication configuration produce improvements that survive selection-robust confirmation. RYS-style free gains do not transfer to vision-transformer representation quality at the scales tested. The structure of the negative result is the paper's positive contribution:

- **Depth-structured tolerance, inverted relative to LLMs.** Early-layer duplication is destructive in every dataset and both models (regional means -3 to $-19\,\mathrm{pp}$); mid-stack duplication is nearly free; late-stack duplication is essentially undetectable ($|\Delta| < 0.4\,\mathrm{pp}$), whereas in LLMs late-layer duplication is catastrophic.
- **The content band is an objective effect, not an architecture effect.** The anatomy experiment finds *no* content-dominant layer band anywhere in EVA-CLIP-18B: rendering style dominates its representational geometry at all 48 layers. In DINOv3-7B a content-dominant band exists (layers 35–40 under CLS pooling; 28–40 under patch pooling)—and it extends to the *final* layer, consistent with the missing-decode-phase account of late-layer immunity.
- **Small but real second-pass effects.** Top configurations in both models reproduce small positive deltas on held-out images with bootstrap confidence intervals excluding zero ($+1.2$ to $+2.0\,\mathrm{pp}$ in EVA-CLIP-18B's late-middle stack; $+0.5$ to $+0.8\,\mathrm{pp}$ in DINOv3-7B's mid-stack), yet remain within what arbitrary-configuration luck can produce under the pre-specified control-band test.

- **A quantitative robustness gap between training regimes.** Averaged over all configurations and datasets, duplication costs EVA-CLIP-18B $-5.8\,\text{pp}$ but DINOv3-7B only $-0.7\,\text{pp}$: the self-supervised model is roughly an order of magnitude more tolerant of depth recycling.

All code, figures, per-image outcomes, and the complete decision log are available at `https://github.com/avchadha/rys`.

2 Background and Related Work

Layer duplication in LLMs. Hankins [2025a] introduced the configuration space studied here: for a network with layers $0, \ldots, L{-}1$, a configuration (i, j) executes layers $0..j{-}1$, feeds the hidden state after layer $j{-}1$ back into layer i, re-executes layers $i..j{-}1$, and continues with layers $j..L{-}1$. On Qwen2-72B, the optimal seven-layer block $(45, 52)$ raised the Open LLM Leaderboard average by $+2.61\,\text{pp}$. Hankins [2025b] replicated the phenomenon across families, introduced single-layer k-fold repetition, and framed results as a quality/compute Pareto frontier; Hankins [2025c] provided the representational account. Depth up-scaling [Kim et al., 2024] exploits related structure with subsequent fine-tuning; we study the pure inference-time intervention.

Layer-level robustness and interpretability. Networks tolerate substantial layer-level perturbation: stochastic depth trains CNNs with randomly dropped layers [Huang et al., 2016]; LayerDrop does the same for transformers [Fan et al., 2020]; Lad et al. [2024] find LLMs robust to deleting or swapping adjacent middle layers, and propose a stages-of-inference picture consistent with the anatomy used here. Intermediate-representation probing goes back at least to linear classifier probes [Alain and Bengio, 2017] and the logit lens [nostalgebraist, 2020]; the cross-lingual variant we adapt [Hankins, 2025c] is closely related to findings that multilingual LLMs route computation through a shared latent space [Wendler et al., 2024]. In vision, Raghu et al. [2021] characterize depth-wise representation structure in ViTs; our anatomy experiment adds a content/style factorization to that program.

Models. EVA-CLIP-18B [Sun et al., 2024] is, to our knowledge, the largest openly released pure vision transformer (17.5B-parameter vision tower; contrastive language supervision). DINOv3-7B [Siméoni et al., 2025] is the largest openly released self-supervised vision transformer (6.7B parameters, image-only distillation with Gram anchoring). Google's ViT-22B [Dehghani et al., 2023] is larger but not publicly released. Vision-language assistants were excluded deliberately: their vision towers are small (e.g., Qwen2.5-VL couples a ~0.7B ViT to LLMs up to 72B), and measuring duplication through a downstream LLM confounds the representational effect with the LLM's tolerance to perturbed inputs (Appendix A.3).

3 Method

3.1 The duplication operator and exhaustive scan

Let h_k denote the hidden state after block k of an L-block encoder. Configuration (i, j), $0 \le i < j \le L$, computes the standard forward pass to h_{j-1},[2] re-enters block i with h_{j-1} as input, re-executes blocks $i..j{-}1$, and continues normally. No weights, normalization statistics, or position information are modified. We scan *all* $L(L{+}1)/2$ configurations (1,176 for EVA-CLIP-18B; 820 for DINOv3-7B), avoiding any grid or sampling artifacts: any contiguous structure observed in the resulting maps is real. A cached implementation makes this tractable: one baseline pass per image batch caches all $L{+}1$ intermediate states, after which configuration (i, j) costs only $(j{-}i) + (L{-}j)$ block evaluations. Equality of the cached and direct paths is verified in unit tests and against each real checkpoint before any scan. We additionally scan single-layer k-fold repetitions ($k \in \{3, 4\}$, exploratory; §4.4).

Features are read from each model's *trained output interface*: the post-layer-norm CLS token for EVA-CLIP-18B, and the final-norm CLS token for DINOv3-7B—applied identically to baseline and duplicated passes, so deltas are attributable solely to the intervention.

[2] Throughout, block indices are zero-based; j is exclusive, so (i, j) duplicates the $j{-}i$ blocks $i, \ldots, j{-}1$.

3.2 Task battery

Following the logic of the original study—whose probes (mathematics and emotional inference) were chosen to engage *different* putative circuits—we evaluate eleven datasets spanning distinct visual domains and distinct visual computations. Six are real-world: ImageNet [Deng et al., 2009] (100-class subset), DTD textures [Cimpoi et al., 2014], EuroSAT satellite imagery [Helber et al., 2019], Places365 scenes [Zhou et al., 2018] (100-class subset), Stanford 40 actions [Yao et al., 2011], and FGVC-Aircraft [Maji et al., 2013]. Five are procedurally generated visual-reasoning probes, each isolating one computation with explicit confound controls (Appendix A): *counting* (numerosity, 10 classes; non-overlapping dots with count-independent radius distributions), *same/different* (relational identity; area-matched distractors), *spatial relation* (4-way relative position; jittered reference), *symmetry* (mirror detection; matched per-half color/size/position marginals), and *inside/outside* (topology; boundary-margin and annulus constraints).

3.3 Probing and debiased evaluation

Each configuration is scored by *training-free* nearest-centroid classification with cosine similarity: class centroids are means of reference-image embeddings, recomputed *within each configuration's feature space*. Unlike a trained probe, this readout cannot adapt to (and thereby mask) representation degradation [Alain and Bengio, 2017]. Reference budgets scale inversely with class count ($n_{\mathrm{ref}}/\mathrm{class} = \mathrm{clip}(\lceil 150/C \rceil, 5, 50)$), equalizing centroid reliability across datasets.

Two test splits. A pilot analysis showed that the natural choice—testing on the images the baseline model finds *hardest*—guarantees spurious positive deltas: with baseline accuracy pinned near zero on an adversarially selected set, *any* perturbation, including pure noise, regresses toward the mean. We instead draw, per dataset, (i) a *borderline* split of 100 images stratified around the decision boundary (50 barely misclassified, 50 barely correct, by |cosine margin|), on which baseline accuracy is $\approx 50\%$ by construction and deltas can move in both directions; and (ii) a *random* control split of 100 images, disjoint from the borderline split, whose deltas are unbiased. **All rankings and headline statistics use the random split**; the borderline split is reported to quantify selection sensitivity (Appendix B.1).

Noise null. For each dataset we measure the Δaccuracy distribution produced by adding isotropic Gaussian noise of relative magnitude $\alpha \in \{0.005, \ldots, 0.1\}$ to both reference and test features (20 seeds each). This weak lower-bound null never exceeded $+0.9\,\mathrm{pp}$ on the random split for EVA-CLIP-18B ($+2.5\,\mathrm{pp}$ for DINOv3-7B's near-ceiling same/different), confirming that the debiased splits do not manufacture positive effects.

Uncertainty. Per-image correctness and true-class rank are persisted for every (configuration, dataset, split); confidence intervals are paired bootstrap over images within dataset (2,000 resamples, datasets treated as fixed effects), with configuration and baseline evaluated on identical resamples.

3.4 Two-stage selection-robust confirmation

With $\sim$1,000 configurations evaluated on 100-image splits (per-cell standard error $\approx$ 3–5 pp), the expected maximum under the global null is $+$15–20 pp: sweep-stage leaders are inflated by construction. Stage two therefore re-evaluates the top 20 configurations (ranked by z-scored random-split aggregate; Appendix B.2) *plus 20 random control configurations* on 500 fresh test images per dataset (disjoint from the entire stage-one candidate pool) with resampled reference sets. The pre-specified confirmation criterion, fixed before either sweep ran, is

$$\mathrm{CI}_{95}(\Delta) > 0 \quad \wedge \quad \Delta > \mu_{\mathrm{ctrl}} + 2\sigma_{\mathrm{ctrl}}, \tag{1}$$

i.e., a configuration must beat both zero and the empirical distribution of arbitrary duplications evaluated under identical conditions. This is the analogue of Hankins [2025b]'s "small probes guide search; large probes judge candidates."

3.5 Representational anatomy with registered predictions

Hankins [2025c] measured, at every layer of an LLM, centered pairwise cosine similarities over an 8×8 topic-by-language sentence grid, finding a middle region where same-topic/different-language

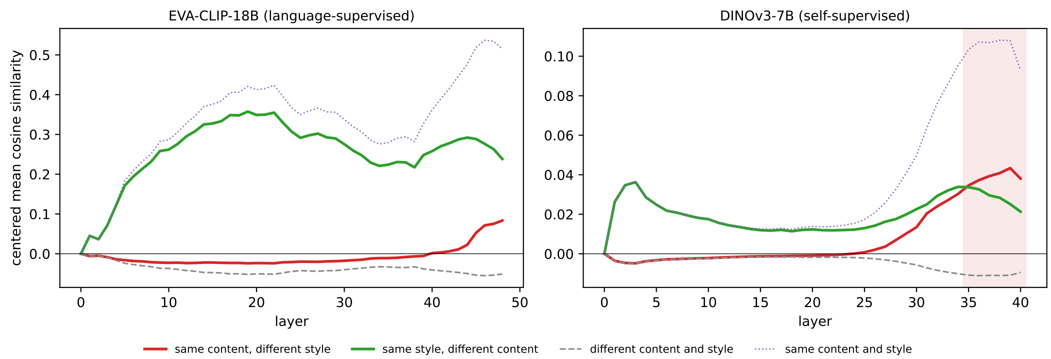


Figure 1: **Layer anatomy under CLS pooling** for the language-supervised (left) and self-supervised (right) model. Curves show per-layer-centered mean pairwise cosine similarity for pairs sharing content (shape) but not style, style but not content, neither, or both (ceiling). In EVA-CLIP-18B, style organization dominates at *every* layer (final-layer content-minus-style gap -0.155). In DINOv3-7B, content overtakes style in a band (shaded) covering layers 35–40—extending to the network's final layer, unlike the LLM case where the band closes $\sim$15 layers before the output.

pairs are more similar than same-language/different-topic pairs—a content-dominant band that coincides with duplication tolerance. Our vision translation replaces language with *rendering style* and topic with *shape identity*: eight shape classes rendered in eight styles (solid, outline, inverted, stippled, striped, sketch, on-black, cluttered background), two jittered instances per cell (128 images; Appendix C). At every layer we pool token representations (CLS primary; patch-mean secondary[3]), center pairwise cosine similarities per layer, and average within pair categories. The *content band* is the set of layers where the same-content curve exceeds the same-style curve.

Critically, the anatomy runs *before* each model's sweep completes, and its predictions were recorded in the project's methods log with timestamps: for EVA-CLIP-18B (no band found), *duplication should be broadly harmful with no beneficial band*; for DINOv3-7B (band at layers 28–40), *tolerance and any gains should concentrate in that band, with early layers fragile*. Section 4.5 grades both.

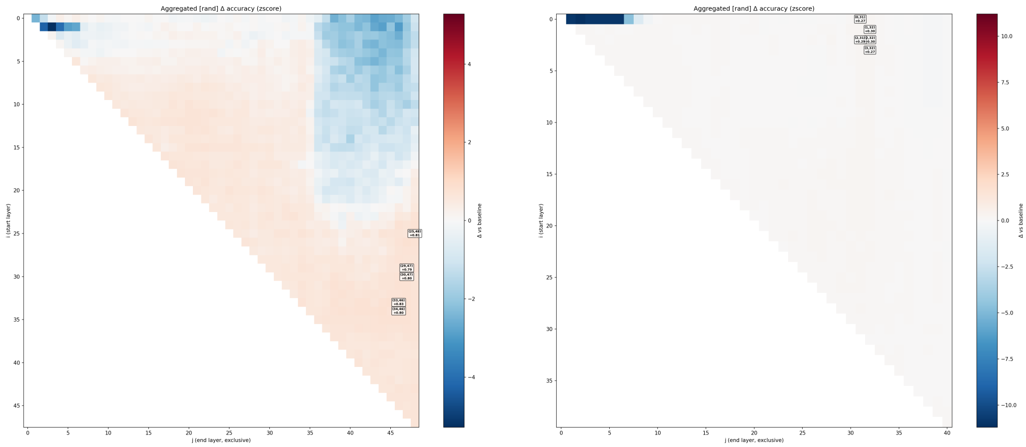
4 Results

4.1 Anatomy: the content band is a training-objective effect

Figure 1 shows the central mechanistic result. In EVA-CLIP-18B, the same-style curve exceeds the same-content curve at all 48 layers: at no depth does the model's representational geometry organize the stimulus grid primarily by shape identity. The content-minus-style gap is maximally 0.000 (at the input embedding, where the CLS token is input-independent) and -0.155 at the final layer. Under patch-mean pooling the gap rises monotonically with depth from -0.69 (background-dominated) to -0.03 at layer 45 without ever crossing zero. In DINOv3-7B, by contrast, the content curve overtakes style at layer 35 and remains dominant through the final layer 40 (CLS pooling; maximum gap $+0.018$; final-layer gap $+0.017$); patch pooling widens the band to layers 28–40.

Two conclusions follow. First, *the absence of a content band in EVA-CLIP-18B is attributable to language supervision, not to vision-transformer architecture*: architecture, stimuli, and analysis are held fixed between the two models. This is consistent with the observation that for a caption-matching objective, rendering style and background *are* content—a caption of a white star on purple mentions the purple—and echoes, at the level of representational geometry, known appearance biases of contrastively trained vision models [Geirhos et al., 2019]. Second, DINOv3-7B's band runs to the final layer. In LLMs the content band closes well before the output because the network must

[3]The style axis mostly manipulates backgrounds, which dominate a patch-mean over $\sim$75% of the canvas; CLS is both the model's own learned summary and the exact vector used by the duplication probes. Patch pooling excludes each model's prefix tokens (one CLS for EVA-CLIP-18B; CLS plus four registers [Darcet et al., 2024] for DINOv3-7B).



(a) EVA-CLIP-18B (48 layers, 1,176 configurations) (b) DINOv3-7B (40 layers, 820 configurations)

Figure 2: **Aggregated duplication maps** (z-scored Δaccuracy on the random split, averaged over eleven datasets; red = improvement, blue = degradation). Row i = first duplicated block; column j = end of duplicated block (exclusive). Both models show the same signature: destructive early rows, a broad harmful region for configurations that begin early and extend deep, and a benign lower-right (late-start) region carrying faint positive structure.

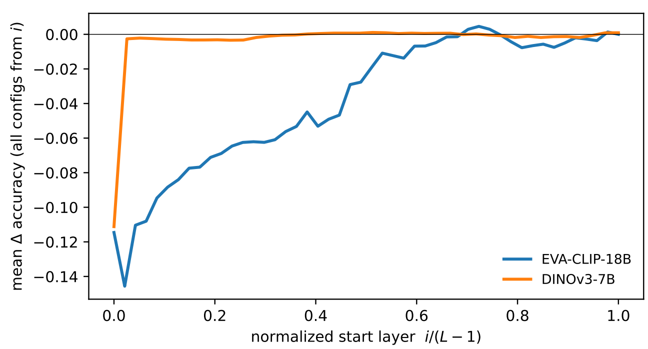


Figure 3: **Depth profile of duplication damage.** Mean random-split Δaccuracy of all configurations starting at layer i, averaged over datasets, against normalized depth. Both models converge to zero damage as the duplicated region moves toward the output—the reverse of the LLM profile, where late-layer duplication is catastrophic. DINOv3-7B is uniformly more tolerant.

re-enter token space [Hankins, 2025c]; an encoder whose output *is* the embedding has nowhere to return to. This is the registered no-decode-phase signature, and it predicts the late-layer duplication immunity observed next. We note the gap magnitudes ($\sim$0.02) are far smaller than their LLM counterparts: style remains strongly represented even in DINOv3-7B.

4.2 Exhaustive sweeps: depth-structured tolerance, inverted relative to LLMs

Figures 2 and 3 and Table 1 summarize 21,956 configuration–dataset evaluations. Three regularities hold in *all* 22 dataset–model combinations:

(1) Early layers are duplication-fragile. Regional means range from -2.9 to -18.8 pp; the damage is largest for texture recognition (DTD, -18.8 pp in EVA-CLIP-18B)—precisely the task most dependent on the low-level statistics early layers compute—and compounds with repetition count (§4.4). Duplicating the patch-adjacent machinery feeds it inputs from the wrong distribution, exactly as the encode-phase account predicts.

(2) Late layers are duplication-immune. Late-region means are statistically indistinguishable from zero in both models (largest magnitude -1.3 pp; DINOv3-7B's late region is $+0.0$ to $+0.06$ pp on

Table 1: **Regional sweep summary** (random split; Δaccuracy in percentage points). Early: $i < 0.15L$, blocks shorter than $0.15L$; mid: $i \in [L/3, 2L/3)$, blocks up to $L/4$; late: $i \geq 0.8L$. Mean/max are over all valid configurations. Per-dataset values in Appendix D.

	EVA-CLIP-18B		DINOv3-7B	
Region	mean	worst dataset	mean	worst dataset
Early	-11.5	-18.8 (DTD)	-10.4	-16.4 (ImageNet)
Mid	-0.9	-4.5 (same/diff.)	$+0.2$	-0.03 (spatial)
Late	-0.6	-3.8 (same/diff.)	-0.1	-1.3 (counting)
All-config mean	-5.8		-0.7	
Best single cell	$+18.0$	(spatial)	$+8.0$	(counting)
Fraction of cells > 0	11.2%		8.1%	

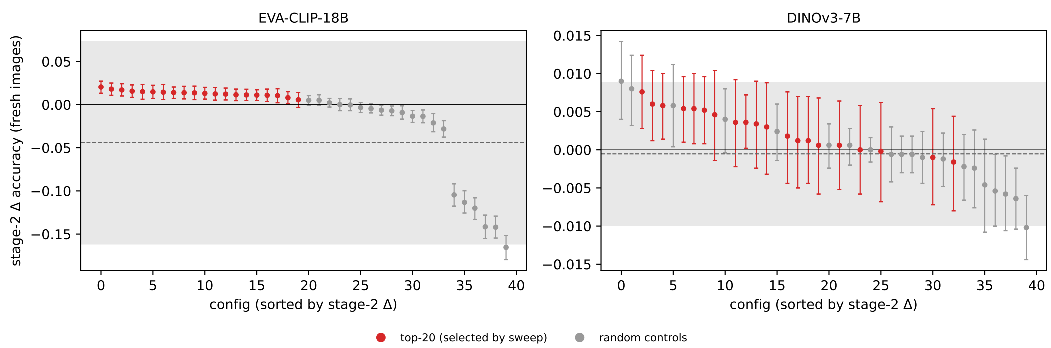


Figure 4: **Stage-two confirmation on fresh images.** Each point is a configuration's across-dataset mean Δaccuracy on 500 held-out images per dataset (bars: 95% paired-bootstrap CIs); red = top-20 from the sweep, gray = random controls; shaded band = $\mu_{\text{ctrl}} \pm 2\sigma_{\text{ctrl}}$. In both models the selected configurations replicate small positive effects (CIs above zero) but none clears the pre-specified control band (Eq. 1).

seven of eleven datasets). This is the mirror image of the LLM result, where duplicating final layers destroys generation, and it is the behavioral counterpart of the anatomy finding that DINOv3-7B's content band extends to its final layer. Notably EVA-CLIP-18B shows the same immunity *without* a content band—late-layer tolerance appears to be a property of encoders as such, not of content-dominant geometry.

(3) Nothing large is gained anywhere. The best single cell in either sweep is $+18$ pp (EVA-CLIP-18B, spatial relations, a coherent neighborhood around $(25-27, 46-47)$)—but at n=100 this is within reach of the winner's curse, which is what stage two is for.

The global tolerance gap is striking: averaged over every configuration and dataset, duplication costs EVA-CLIP-18B -5.8 pp and DINOv3-7B -0.7 pp. The self-supervised model's representations pass through a doubled mid-or-late stack nearly unchanged. We attribute part of this to DINOv3-7B's near-ceiling baselines on several probes (its features solve inside/outside at 99.4% and symmetry at 99.1%, versus 83.9% and 100.0% for EVA-CLIP-18B; Appendix D), but the gap persists on datasets with ample headroom (counting: -0.3 vs. -5.4 pp; DTD: -0.9 vs. -5.6 pp; FGVC: -0.4 vs. -10.0 pp).

4.3 Stage-two confirmation: no configuration survives

Figure 4 shows the confirmation outcome; full tables are in Appendix 5. For EVA-CLIP-18B, the twenty random controls average $\mu_{\text{ctrl}} = -4.4$ pp ($\sigma_{\text{ctrl}} = 5.9$ pp)—arbitrary duplication hurts—so the criterion band sits at $+7.3$ pp. The top-ranked configurations reproduce positive deltas on fresh images—all twelve leaders have 95% CIs excluding zero, led by $(34, 41)$ at $+2.0$ pp $[+1.3, +2.7]$—but none approaches the band. For DINOv3-7B, controls are tightly concentrated ($\mu_{\text{ctrl}} = -0.05$ pp, $\sigma_{\text{ctrl}} = 0.47$ pp; random duplication barely hurts this model), the band sits at $+0.9$ pp, and again

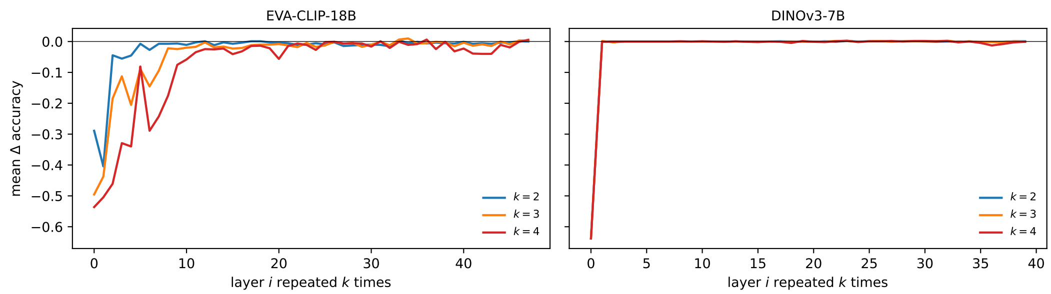


Figure 5: **Single-layer repetition** (exploratory; no confirmation stage). Mean random-split Δaccuracy over datasets when layer i is executed k times. Early-layer damage compounds with k in EVA-CLIP-18B (regional mean -22 pp at k=3, -35 pp at k=4) but saturates in DINOv3-7B; mid- and late-stack repetition remains near-free in both models even at k=4.

Table 2: **Registered predictions versus outcomes.** Predictions were recorded in the project log before the corresponding sweep results existed.

Registered prediction	Outcome
EVA-CLIP-18B: no content band ⇒ duplication broadly harmful, no beneficial band	✓ Directionally correct (all regional means $\leq$ 0; nothing confirmed), but under-specified: the harm is strongly depth-structured rather than uniform
DINOv3-7B: early layers fragile	✓ Confirmed, all datasets
DINOv3-7B: duplication tolerated within the band (layers 28–40)	✓ Band deltas $\approx$ 0.000; but tolerance is not band-exclusive—the mid-stack is equally immune, so the band *under-predicts* the tolerant region
DINOv3-7B: positive effects concentrated in/near the band	✗ The (small, unconfirmed) positive effects concentrate at $i \approx 13-21$, largely *below* the band

no selected configuration clears it; the best is $(14, 30)$ at $+0.76$ pp $[+0.28, +1.24]$. One control configuration exceeds the band ($+0.90$ pp), consistent with the ≈ 0.5 false positives expected among twenty controls at the 2σ level.

Verdict under Eq. 1: zero confirmed winners in either model. We emphasize the two-sided reading. The pre-specified test asks whether selected configurations beat what *arbitrary* duplication achieves by luck, and they do not—so claims of RYS-style circuit gains are not supported. At the same time, the replication of small positive deltas on held-out data with resampled references is itself selection-free evidence of a weak, real second-pass benefit, concentrated in EVA-CLIP-18B's late-middle stack (intriguingly, the leader $(34, 41)$ is a seven-layer block, the same width as the original RYS optimum on Qwen2-72B) and DINOv3-7B's mid-stack. The effect sizes ($+0.5$–2 pp) are an order of magnitude below the LLM gains that motivated the study.

4.4 Single-layer k-fold repetition

RYS-II reported that repeating a single well-chosen LLM layer three times yields cheap gains [Hankins, 2025b]. We find no vision analogue (Figure 5): single-layer repetition is neutral at best. Its depth profile, however, sharpens the sweep's picture. In EVA-CLIP-18B, early-layer damage compounds nearly multiplicatively with k (early-region mean -22.1 pp at k=3, -34.8 pp at k=4), while DINOv3-7B's early-layer damage saturates (-10.7 pp at both k) and its mid-stack is astonishingly inert ($+0.01$ pp at k=3 *and* k=4): a DINOv3-7B middle layer can be run four times in a row with no measurable representational consequence.

4.5 Registered-prediction scorecard

Table 2 grades the pre-registered predictions. Content-dominant geometry correctly identifies *where duplication is safe* but neither exhausts the tolerant region nor localizes the (weak) benefits. Representational stability of the kind measured by the anatomy is evidently necessary-but-coarse: a sufficient account of duplication tolerance must appeal to more than the content/style organization of the residual stream.

5 Discussion

Why no reasoning circuits in vision? The most economical reading of our negative result combines three observations. First, the LLM gains arose in *generative, multi-step* inference, where a duplicated block participates repeatedly in an autoregressive loop and small per-step refinements can compound; a single-pass encoder offers one opportunity, and we measure its output geometry directly. Second, the anatomy shows that the representational precondition identified in LLMs—a wide format-agnostic band—is absent in the language-supervised model and thin (six layers, weak gaps) even in the self-supervised one. Third, where a mild version of the precondition holds (DINOv3-7B), we observe exactly a mild version of the LLM phenomenon: perfect tolerance and sub-percentage-point, CI-positive second-pass effects. Vision transformers may simply sit lower on the same curve, their single-pass objectives providing little pressure to develop computations that profit from re-execution.

The missing decode phase. The cleanest transferable structure is the inversion of late-layer behavior. In LLMs the final layers perform format-specific work whose duplication is catastrophic; in both our models late-layer duplication is free, and in DINOv3-7B the content band runs to the final layer. Encoders end *in* representation space. This has a practical corollary: inference-time depth interventions in vision encoders (adaptive compute, speculative shallow paths, layer skipping [Fan et al., 2020, Lad et al., 2024]) can treat the late stack as a safe region, in sharp contrast to LLM practice.

Objective, not architecture. The two arms differ only in checkpoint; the anatomy differs qualitatively. Language supervision, on this evidence, prevents the formation of (or masks) appearance-invariant layer geometry: contrastive image-text training keeps style information linearly prominent to the last layer, presumably because captions describe appearance. Self-supervision alone produces the abstraction gradient. This offers a representational-geometry complement to behavioral findings on appearance bias [Geirhos et al., 2019] and a caution for mechanistic claims that generalize across training regimes.

Methodological note. The gap between our sweep-stage leaders ($+18\,\mathrm{pp}$) and confirmed effects ($\leq +2\,\mathrm{pp}$, below the control band) quantifies, in a real experiment, how much of an exhaustive-search leaderboard is selection noise. We commend the two-stage protocol with random-configuration controls—and the practice of registering mechanistic predictions before outcome data exist—to future work on inference-time interventions, where the configuration spaces are large and the evaluations necessarily small.

6 Limitations

(1) We probe *representation quality* at the encoder's trained output interface, not downstream behavior; the original RYS effects were behavioral. A duplication that improves, e.g., VQA through a vision-language pipeline without improving nearest-centroid geometry would escape our measurement (a natural follow-up; Appendix A.3). (2) Datasets are treated as fixed effects and several probes are at or near ceiling for DINOv3-7B, muting detectable upside there. (3) Two synthetic probes retain documented residual cues (inside/outside distance statistics; spatial-relation absolute position), and one (symmetry) is at ceiling for both models. (4) The anatomy stimuli are procedural; for a contrastive model, style is arguably legitimate content, which our objective-ablation design addresses but does not eliminate. (5) Two models, one run each (deterministic inference); conclusions are about these checkpoints and scales.

7 Conclusion

We performed the first exhaustive layer-duplication study in vision transformers, at the largest openly available scales, under a selection-robust, pre-registered protocol. Inference-time layer duplication does not buy vision what it bought language models: no configuration among 1,996 produces confirmed gains in either a language-supervised or a self-supervised frontier ViT. What the intervention reveals instead is structure: a universal early-layer fragility, a late-layer immunity that inverts the LLM picture and follows from encoders' lack of a decode phase, a content-dominant band that exists only without language supervision, and a small but replicable second-pass benefit that hints the LLM phenomenon has a faint visual shadow. All artifacts required to reproduce or extend every number and figure in this paper are public.

Reproducibility statement

Code, the complete decision log (`docs/methods.md`), all score matrices, per-image outcomes, probe-set indices, anatomy features, and generated figures are available at `https://github.com/avchadha/rys` (raw data on the `results-sync` branch). Every stochastic choice is seeded; dataset stimuli for the synthetic probes and anatomy grid are regenerable from code. Hardware and cost: one NVIDIA H100 80GB (Lambda Cloud); $\sim$27.5 h for the EVA-CLIP-18B arm, $\sim$6 h for the DINOv3-7B arm, $\approx$\$95 total. Software pins and verified checkpoint-interface facts are listed in Appendix F.

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A Task battery details

A.1 Real-world datasets

ImageNet-1k and Places365 are subsampled to 100 classes each (deterministic seed) to equalize probe cost; all other datasets use their full label sets (DTD 47, EuroSAT 10, Stanford 40 40, FGVC-Aircraft 100). Images pass through each model's native 224×224 preprocessing.

A.2 Synthetic visual-reasoning probes

All probes are generated procedurally at 448×448 (downscaled by the model transform), with disjoint generation seeds for reference/candidate pools, and are regenerable exactly from code. Each was adversarially reviewed for shortcut cues; the shipped versions include the following controls:

Counting (10 classes; 1–10 dots). Radii drawn i.i.d. $\mathcal{U}\{12..40\}$ *before* placement (count-independent size distribution); positions rejection-sampled for strict non-overlap ($\geq$ 6px gap) with whole-layout restart on failure, so visible count always equals the label. Residual cue: total colored area correlates with count (inherent to numerosity stimuli).

Same/different (2 classes). Two polygons; "same" pairs share the vertex set, translated and rotated ± 0.6 rad; "different" pairs use a fresh polygon with the same vertex count, rescaled to match the reference polygon's area exactly. Colors always differ between the two shapes.

Spatial relation (4 classes). Red reference disc jittered $\pm$70px around center; blue triangle at gap 80–160px with $\pm$25px cross-axis jitter. Residual cue: absolute position retains partial information (jitter is smaller than the maximum gap).

Symmetry (2 classes). Mirror-symmetric versus scrambled dot patterns; scrambled images reuse the mirrored dots' exact (y, radius, color) multiset and resample only x, so per-half color histograms, size distributions, and vertical marginals are identical between classes.

Inside/outside (2 classes). Irregular star-shaped contour (6–12 vertices, radii 0.6–1.0R, $R \in [100, 160]$) drawn as outline; the query dot must clear the boundary by a visible margin and lie in a distance annulus where both classes occur; contours are regenerated when placement is infeasible; ray casting provides ground truth. Residual cue: distance-from-center separates the classes partially (weakened, not eliminated).

A.3 Excluded designs

Vision-language models were excluded because (i) their vision towers are small relative to the models studied here (Qwen2.5-VL uses a $\sim$0.7B-parameter ViT across its 3B–72B family), and (ii) measuring duplication through an LLM confounds representational change with the LLM's tolerance to off-distribution visual tokens. Applying the best/worst configurations found here to a VLM's vision tower and measuring task behavior is the natural behavioral follow-up. CLEVR and additional fine-grained natural-photo datasets were excluded as redundant with existing battery members; beam-searched multi-block compositions and surrogate-model search [Hankins, 2025b] were excluded because the configuration space is exhaustively enumerable at these depths.

B Probe methodology details

B.1 Borderline split construction

For each dataset, 1,000 candidate images are drawn from the test split (restricted to the class subset). Under baseline features, each candidate receives a margin $m = s_{\text{true}} - \max_{c\neq\text{true}} s_c$ (cosine similarities to class centroids). The borderline split comprises the 50 misclassified candidates with margin closest to zero and the 50 correctly classified candidates with margin closest to zero. The random split is drawn uniformly from the remaining candidates. Selecting instead the *most negative* margins (the original plan, discarded before any GPU experiments) pins baseline accuracy near zero and produces guaranteed-positive deltas under any perturbation. The empirical comparison bears this out: on the borderline split, sweep maxima reach $+21$ pp (same/different) versus $+4$ pp on the random split for the same model and dataset—an order-of-magnitude demonstration of selection sensitivity (the borderline-split aggregate map is shown in Figure 7).

B.2 Aggregation

Per-dataset Δ matrices are aggregated two ways: an unweighted mean (interpretable units) and a mean of per-dataset z-scores, where each dataset's deltas are standardized over its own valid cells. The z-scored random-split accuracy matrix is the canonical ranking object; it prevents high-variance binary probes from dominating the mean.

B.3 Noise null

Gaussian noise with per-dimension standard deviation $\alpha\,\overline{\|f\|}/\sqrt{D}$ is added to reference and test features simultaneously, $\alpha \in \{0.005, 0.01, 0.02, 0.05, 0.1\}$, 20 seeds each; the maximum mean Δaccuracy across magnitudes is reported per dataset in Tables 3–4. This null is deliberately weak (duplication shifts features coherently, not isotropically); the matched null is the random-configuration control of the confirmation stage.

B.4 Bootstrap

Per-image correctness bitmaps are persisted for every (configuration, dataset, split). CIs are percentile intervals from 2,000 paired bootstrap resamples over images within each dataset, with per-dataset deltas averaged across datasets in each resample [Efron and Tibshirani, 1994]. Datasets are fixed effects; CIs are conditional on the reference sets (stage two resamples them).

C Anatomy stimulus grid

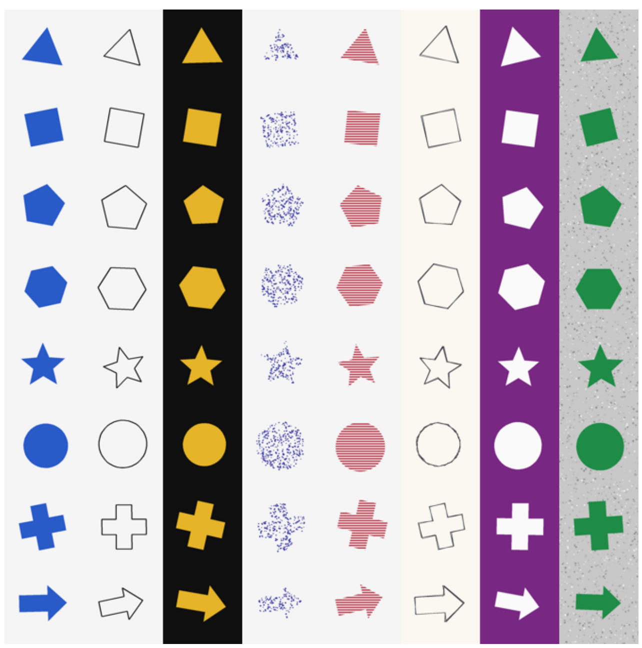


Figure 6: **The** 8×8 **style-by-content grid** (one jitter instance shown per cell). Rows: shape classes (triangle, square, pentagon, hexagon, star, circle, cross, arrow). Columns: rendering styles (solid-on-white, outline, solid-on-black, stippled, striped, sketch, inverted, cluttered background).

Shapes are defined as canonical vertex geometry and rendered at radius $\approx$ 140px on a 448×448 canvas. Per-instance jitter (rotation $\pm15°$, translation ±12px, scale $\times0.85$–1.0) defeats pixel-level matching between same-shape pairs while preserving unambiguous shape identity; larger rotations were rejected because they corrupt content labels (a square at $45°$ is a diamond). Two instances per cell yield 128 images and add a same-content-same-style ceiling category. With category counts 896 / 896 / 6,272 / 64 (same-content, same-style, different, ceiling), the three primary curves average over hundreds of pairs per layer.

D Full result tables

Table 3: **EVA-CLIP-18B per-dataset summary** (random split; accuracies as fractions, deltas in fractions of accuracy). Border/rand: baseline accuracy on the borderline/random split. Noise: maximum mean Δaccuracy under the Gaussian null. Mean/max over all 1,176 configurations; early/mid/late as in Table 1.

Dataset	Cls	Refs	Cand	Border	Rand	Noise	Mean	Max	Early	Mid	Late
counting	10	150	.392	.500	.390	$+.000$	$-.054$	$+.070$	$-.033$	$-.013$	$+.006$
DTD	47	235	.684	.500	.750	$+.006$	$-.056$	$+.030$	$-.188$	$-.001$	$-.004$
EuroSAT	10	150	.899	.500	.930	$+.000$	$-.069$	$+.020$	$-.150$	$-.003$	$-.001$
FGVC-Aircraft	100	500	.691	.500	.730	$+.000$	$-.100$	$+.030$	$-.132$	$-.014$	$-.004$
ImageNet-100	100	500	.925	.500	.960	$+.002$	$-.047$	$+.010$	$-.163$	$-.001$	$-.002$
inside/outside	2	100	.839	.500	.900	$+.009$	$-.062$	$+.060$	$-.029$	$-.017$	$-.002$
Places365-100	100	500	.587	.500	.600	$+.000$	$-.062$	$+.040$	$-.120$	$-.006$	$-.011$
same/different	2	100	.711	.500	.820	$+.007$	$-.102$	$+.040$	$-.128$	$-.045$	$-.038$
spatial relation	4	152	.818	.500	.800	$+.004$	$-.033$	$+.180$	$-.060$	$-.002$	$-.002$
Stanford 40	40	200	.862	.500	.920	$+.008$	$-.042$	$+.040$	$-.111$	$+.000$	$-.004$
symmetry	2	100	1.000	1.000	1.000	$+.000$	$-.016$	$+.000$	$-.048$	$+.000$	$+.000$

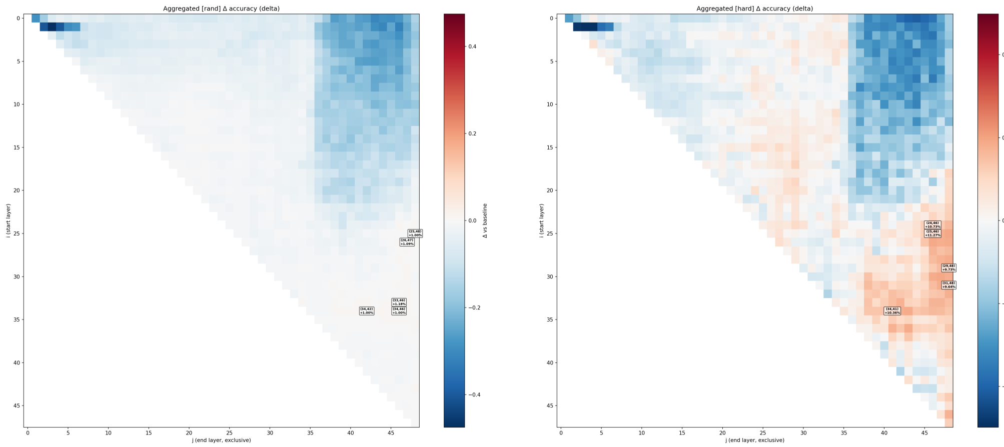
Table 4: **DINOv3-7B per-dataset summary** (columns as in Table 3; 820 configurations). Note the near-ceiling baselines on the geometric probes and ImageNet, and the dramatically smaller whole-matrix means.

Dataset	Cls	Refs	Cand	Border	Rand	Noise	Mean	Max	Early	Mid	Late
counting	10	150	.430	.500	.420	$+.000$	$-.003$	$+.080$	$-.057$	$+.014$	$-.013$
DTD	47	235	.695	.500	.750	$+.000$	$-.009$	$+.020$	$-.118$	$+.004$	$+.001$
EuroSAT	10	150	.891	.500	.950	$+.011$	$-.011$	$+.000$	$-.137$	$-.000$	$+.000$
FGVC-Aircraft	100	500	.492	.500	.450	$+.009$	$-.004$	$+.020$	$-.075$	$+.000$	$-.001$
ImageNet-100	100	500	.935	.500	.990	$-.000$	$-.010$	$+.010$	$-.164$	$+.000$	$+.000$
inside/outside	2	100	.994	.940	1.000	$+.000$	$-.003$	$+.000$	$-.077$	$+.000$	$+.000$
Places365-100	100	500	.555	.500	.610	$+.003$	$-.009$	$+.020$	$-.097$	$+.000$	$+.001$
same/different	2	100	.830	.500	.930	$+.025$	$-.001$	$+.020$	$-.080$	$+.001$	$+.000$
spatial relation	4	152	.938	.500	.990	$-.000$	$-.008$	$+.000$	$-.113$	$-.000$	$+.000$
Stanford 40	40	200	.886	.500	.960	$+.018$	$-.011$	$+.000$	$-.158$	$+.000$	$+.000$
symmetry	2	100	.991	.910	1.000	$+.000$	$-.004$	$+.000$	$-.098$	$+.000$	$+.000$

Table 5: **Stage-two confirmation, leading configurations.** Deltas are across-dataset means on 500 fresh images per dataset with resampled reference sets; brackets: 95% paired-bootstrap CIs. "Beats null": Eq. 1. Controls: EVA-CLIP-18B $\mu = -.0441$, $\sigma = .0586$; DINOv3-7B $\mu = -.0005$, $\sigma = .0047$.

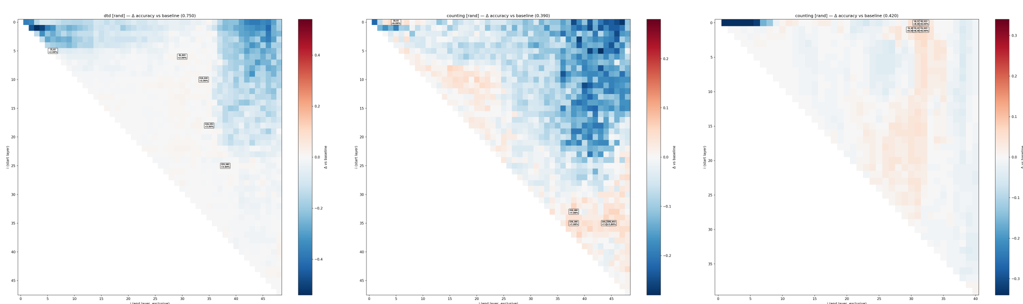
EVA-CLIP-18B				DINOv3-7B			
Config	Δ	CI	Beats	Config	Δ	CI	Beats
(34, 41)	$+.0204$	$[+.0132, +.0272]$	no	(13, 28)^c	$+.0090$	$[+.0040, +.0142]$	yes*
(34, 44)	$+.0180$	$[+.0108, +.0250]$	no	(16, 28)^c	$+.0080$	$[+.0032, +.0124]$	no
(34, 42)	$+.0170$	$[+.0100, +.0242]$	no	(14, 30)	$+.0076$	$[+.0028, +.0124]$	no
(33, 41)	$+.0156$	$[+.0084, +.0228]$	no	(13, 31)	$+.0060$	$[+.0012, +.0104]$	no
(25, 48)	$+.0150$	$[+.0062, +.0234]$	no	(20, 27)	$+.0058$	$[+.0014, +.0100]$	no
(30, 47)	$+.0148$	$[+.0072, +.0224]$	no	(13, 26)^c	$+.0058$	$[+.0004, +.0112]$	no
(24, 48)	$+.0144$	$[+.0060, +.0234]$	no	(21, 27)	$+.0054$	$[+.0010, +.0096]$	no
(34, 46)	$+.0140$	$[+.0072, +.0204]$	no	(15, 31)	$+.0054$	$[+.0008, +.0100]$	no
(33, 44)	$+.0138$	$[+.0066, +.0212]$	no	(16, 31)	$+.0052$	$[+.0008, +.0096]$	no
(29, 47)	$+.0136$	$[+.0058, +.0214]$	no	(4, 31)	$+.0046$	$[-.0014, +.0104]$	no

^{c}random control configuration. *one control exceeding the band matches the expected 2σ false-positive rate among twenty controls (≈ 0.5); no *selected* configuration meets the criterion in either model.



(a) Random split (unbiased). (b) Borderline split (selection-sensitive).

Figure 7: **Selection sensitivity, demonstrated** (EVA-CLIP-18B, aggregated raw Δaccuracy). The borderline split (right)—whose test images were selected by their behavior under the baseline model—paints a substantially rosier picture than the unbiased random split (left) for identical interventions. All conclusions in this paper are drawn from the random split; the pilot design based on *hardest* images (not shown) was more distorted still.



(a) EVA-CLIP-18B, DTD. (b) EVA-CLIP-18B, counting. (c) DINOv3-7B, counting.

Figure 8: **Representative per-dataset duplication maps** (random split, Δaccuracy). Texture recognition (a) shows the strongest early-layer fragility; numerosity (b, c), the probe with the most headroom, shows the mild positive mid/late-stack structure in both models.

E Additional figures

F Implementation, verification, and reproducibility

Checkpoint interface facts (verified against source before any GPU run): EVA-CLIP-18B ships no preprocessor configuration (its model card uses the OpenAI CLIP processor, which we adopt); its encoder blocks take two required positional mask arguments; its vision tower applies a post-layer-norm before pooling, which we include in the probed feature. DINOv3-7B computes rotary position embeddings once per input, shared by all blocks; its four register tokens [Darcet et al., 2024] are excluded from patch pooling; its final norm is included in the probed feature. For both models, our re-implemented forward path reproduces the checkpoint's own `forward` to numerical identity on random-initialization tests, and cached-path/direct-path equality is asserted on the real checkpoints before scanning.

Verification. A CPU-only test suite (25 tests) covers the scanner algebra (cache/direct equivalence, repetition semantics, validation), probe scoring, borderline stratification, every synthetic confound control (non-overlap, area matching, half-marginal identity, boundary margins, label validity), anatomy pair-category combinatorics, recovery of planted structure, and an end-to-end miniature

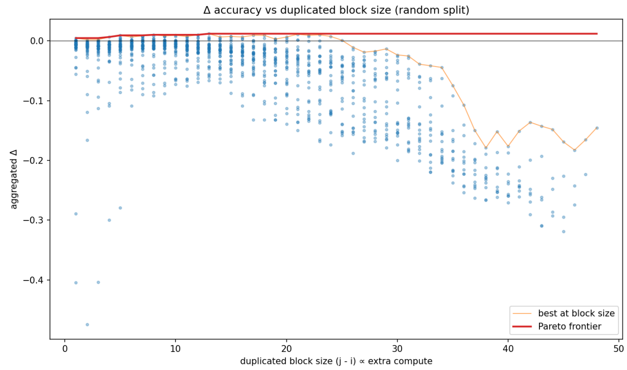
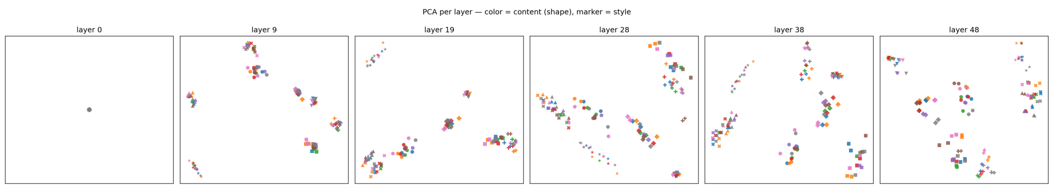
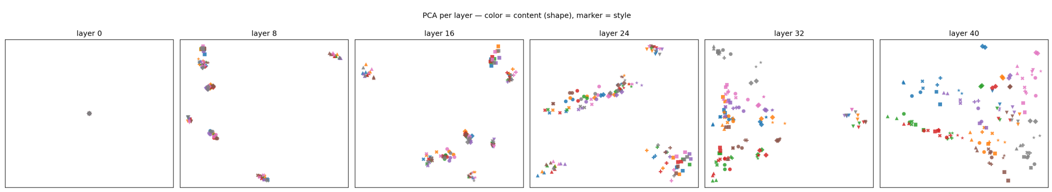


Figure 9: **Quality/compute Pareto view** (EVA-CLIP-18B, random split): aggregated Δaccuracy against duplicated block size ($\propto$ extra compute), with the frontier line. No block size offers a favorable trade at this scale.



(a) EVA-CLIP-18B: cells arrange by style (color mixes within clusters).



(b) DINOv3-7B: late-layer panels cluster by content color.

Figure 10: **Per-layer PCA of anatomy features** (CLS pooling; point color = shape class, marker = rendering style).

pipeline (scan $\rightarrow$ resume $\rightarrow$ figures $\rightarrow$ report $\rightarrow$ confirmation) with a dummy transformer. A simulated mid-scan crash repairs bit-identically under resume.

Determinism. All sampling is seeded (reference/candidate selection, split draws, confirmation images and references, control configurations, synthetic generation, class subsets); models run in evaluation mode in fp16. Residual GPU nondeterminism is below the quantization of 100-image accuracies.

Software. PyTorch 2.13 (CUDA); `transformers` 4.49 for the EVA-CLIP-18B arm (its remote code predates API changes in 4.50+) and 5.14 for the DINOv3-7B arm, in separate environments; `datasets` 2.21 (< 3, required by the script-based ImageNet loader). Complete pins, seeds, and the full decision log accompany the repository.

Scale of the experiment. 21,956 configuration–dataset evaluations (sweeps), 1,936 single-layer repetition evaluations, 880 confirmation evaluations (40 configurations $\times$ 11 datasets $\times$ 2 stages of features), and two anatomy passes, in $\approx$ 33.5 GPU-hours on one H100.